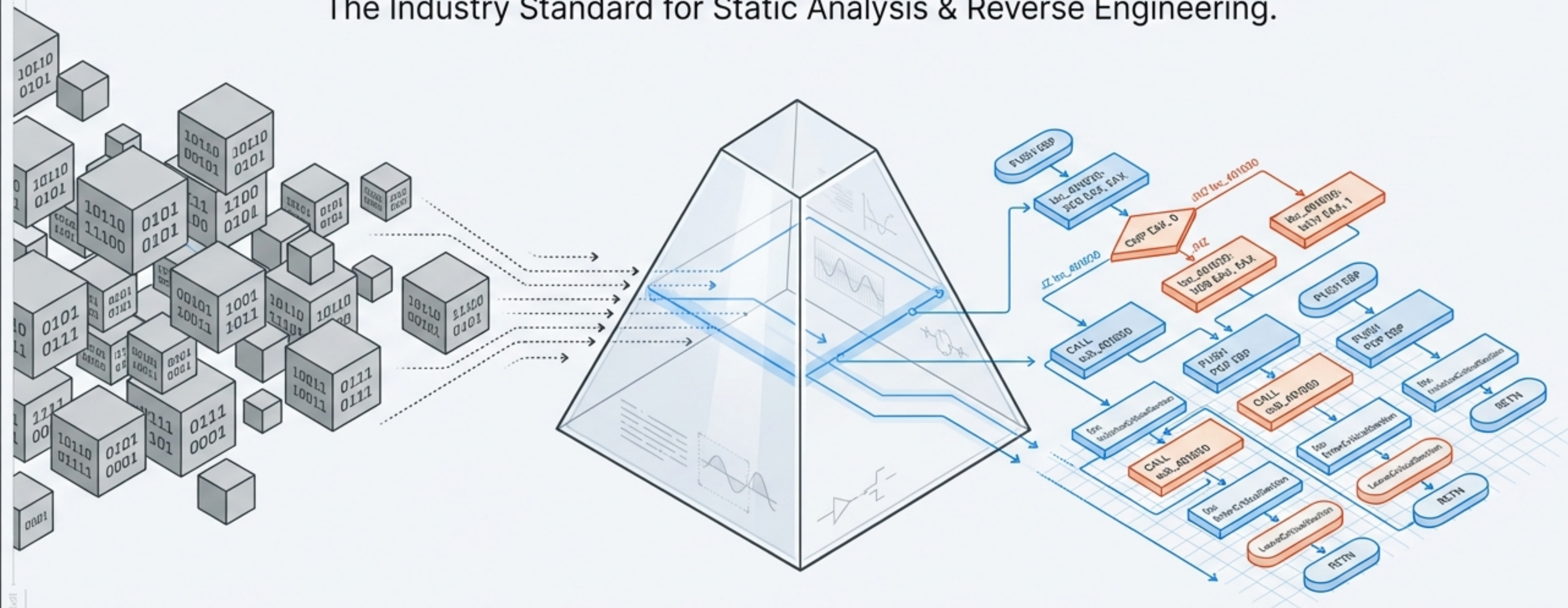


Reconstructing Logic with IDA Pro

The Industry Standard for Static Analysis & Reverse Engineering.



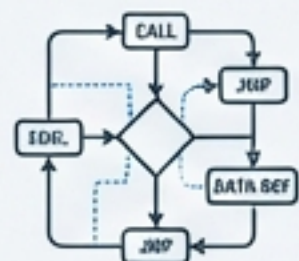
Reverse engineering is not merely reading code; it is the act of reconstructing logic from raw data. IDA Pro (Interactive Disassembler) is the primary tool for this task.

More Than a Translator

IDA is a “multipass” disassembler. While a standard disassembler simply translates machine code to assembly, IDA **analyzes context** to reconstruct the program's original intention.



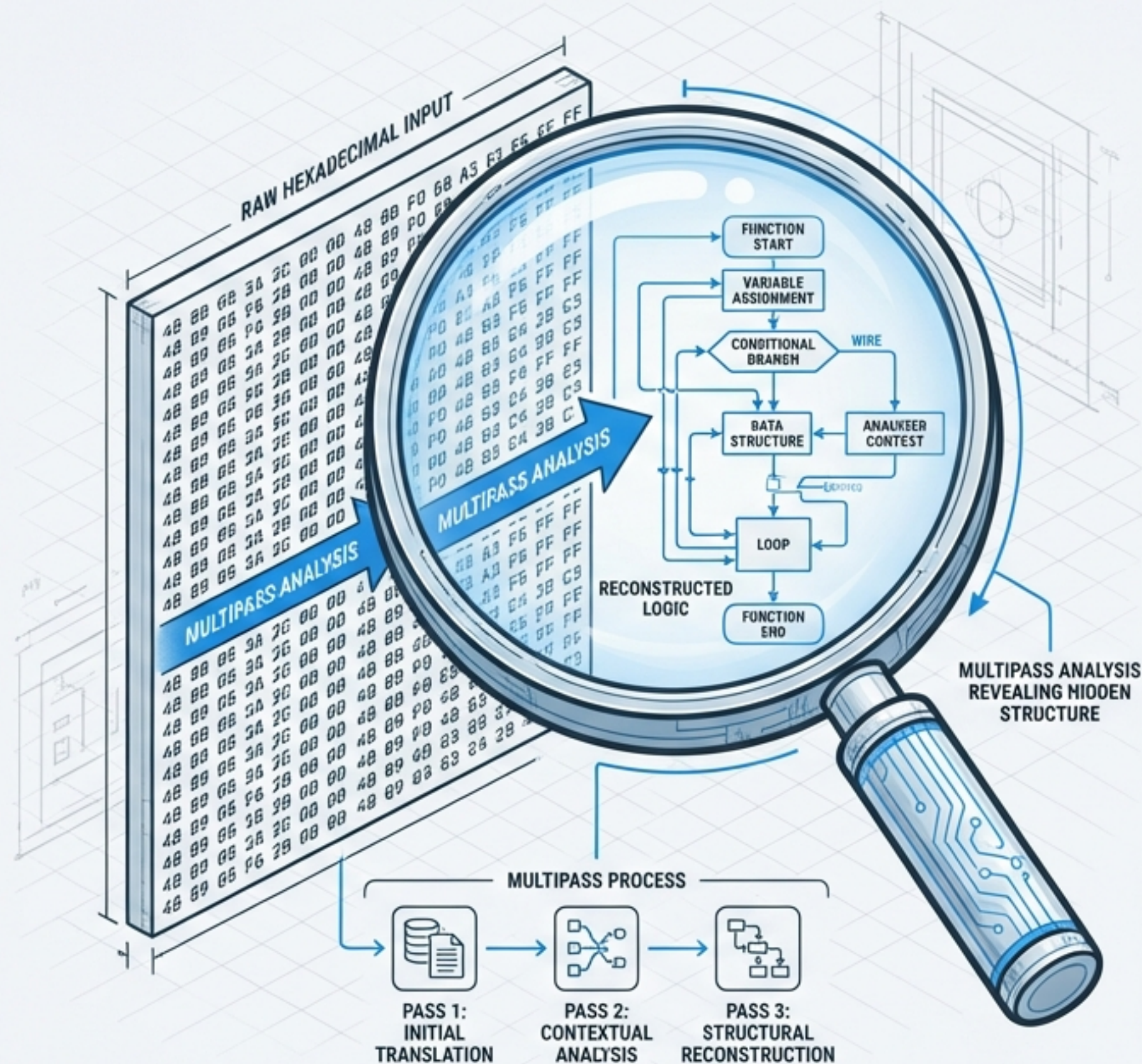
- **Identifies Structures:** Locates functions, variables, and data structures.



- **Maps Relationships:** Specific cross-references (xrefs) show how different parts of the code interact.

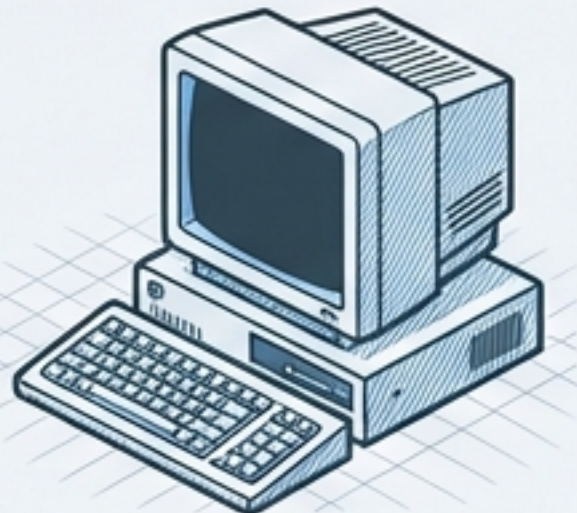


- **Reconstructs Logic:** Converts raw bytes into a navigable map of the software's behavior.



Choose the Right Tool for the Mission

Navigating IDA versions is the first hurdle. For this course, functionality (saving) outweighs aesthetics.

Freeware 5.0	Demo Version	Pro Version
		
Status: REQUIRED The Lab Standard. Old UI, limited x64 support.	Status: PRACTICE ONLY Modern UI, latest processors.	Status: INDUSTRY STANDARD Full feature set.
Allows Saving (.idb)	Saving Disabled	Extremely Expensive
Must use for labs.	Cannot submit work.	Professional use only.

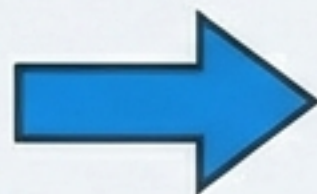
CRITICAL: To submit your labs, you must use Freeware 5.0. The Demo version will result in lost work.

The Handshake: Loading the Binary



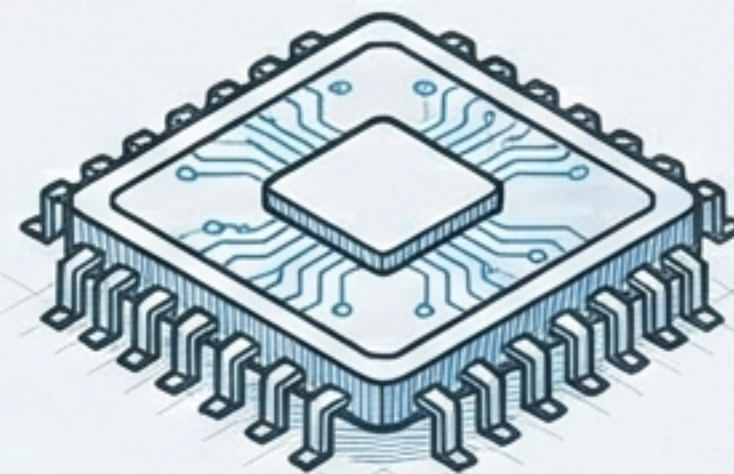
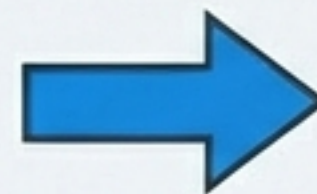
1. Ingest

Drag & Drop or File > Open.



2. Detection

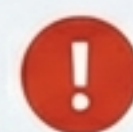
IDA identifies the PE (Portable Executable) format.



3. Architecture

IDA guesses the processor (e.g., Intel 80x86).

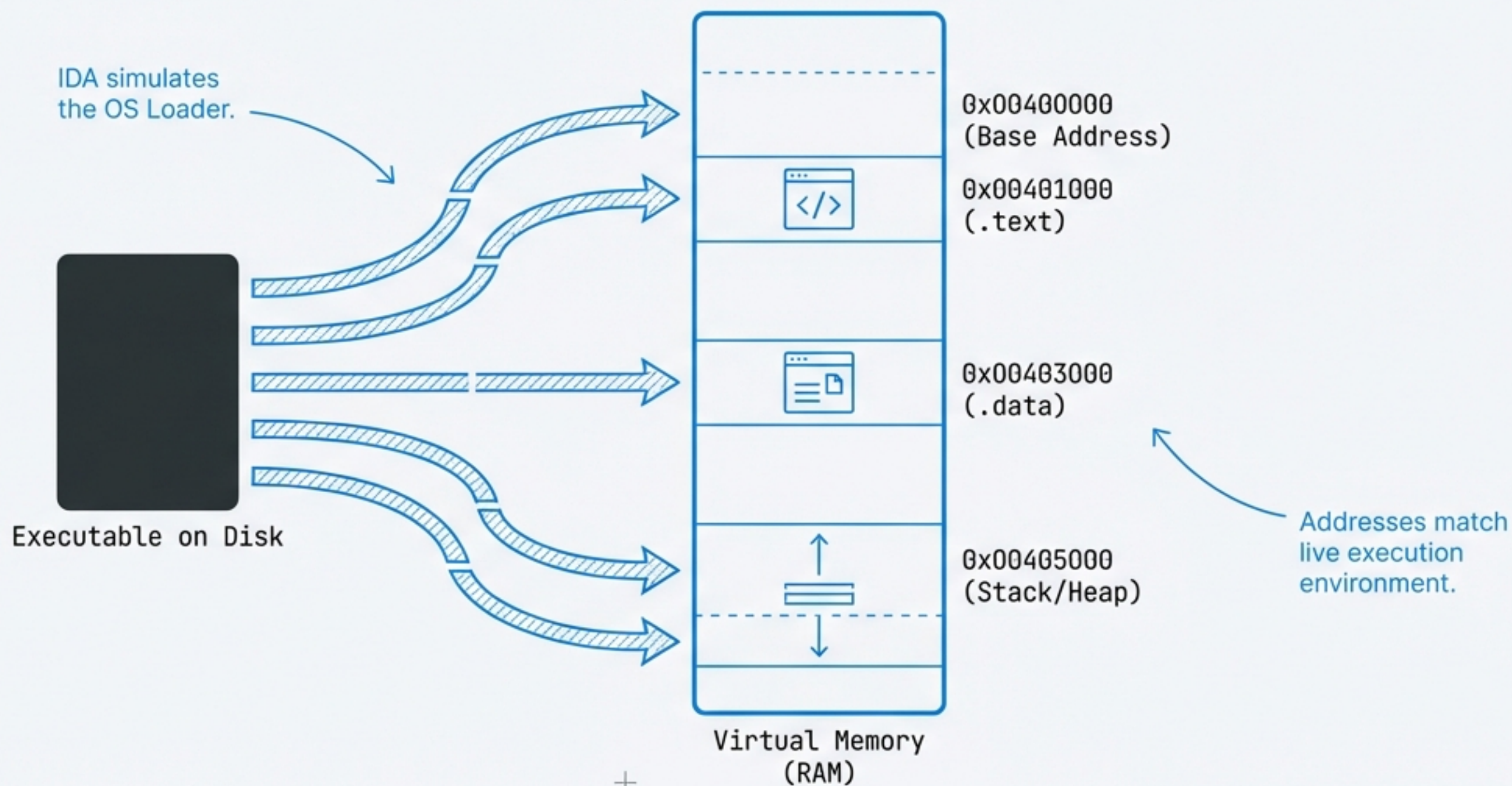
The Golden Rule: Accept the Defaults.
IDA's auto-detection is excellent.



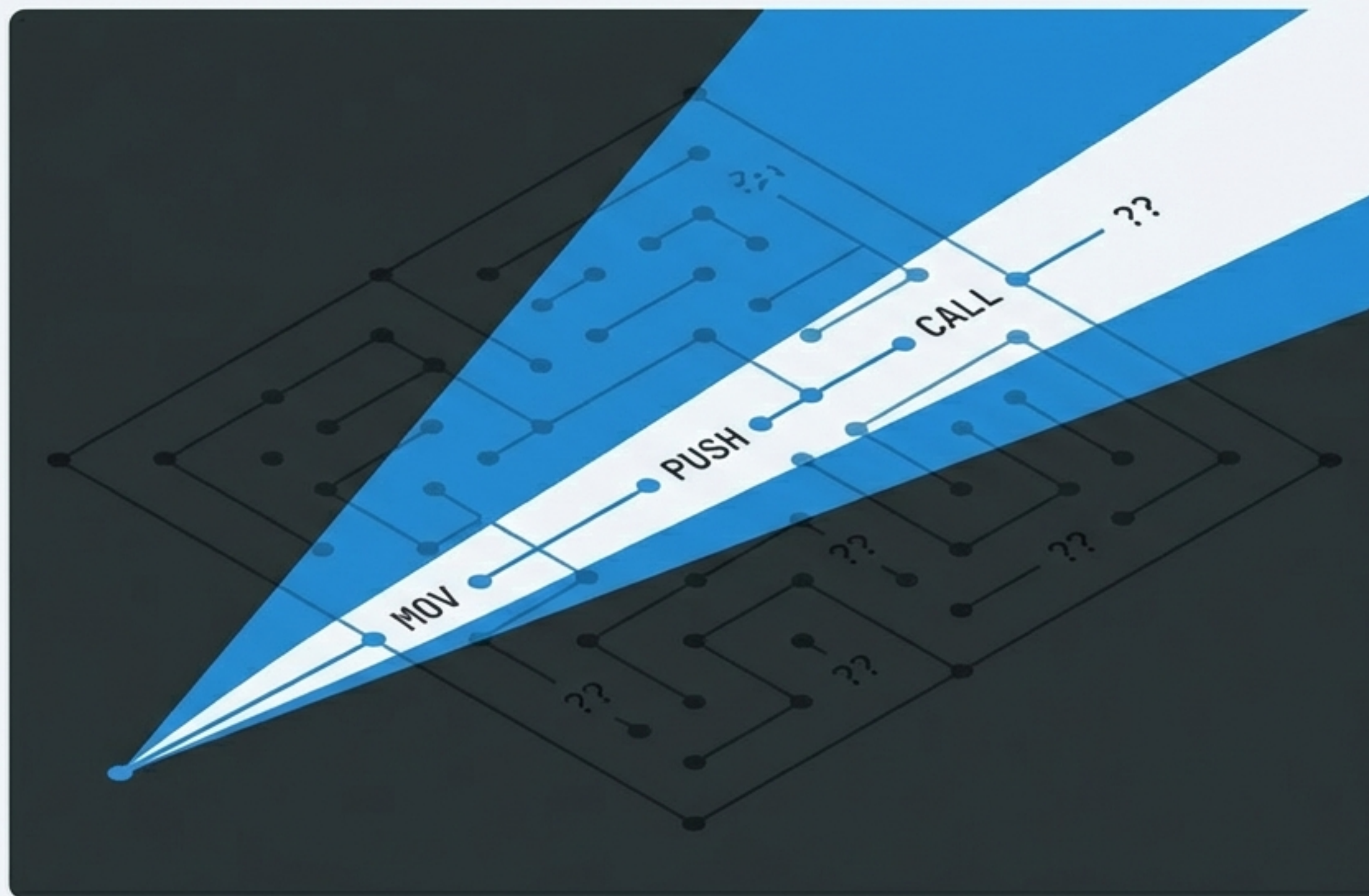
Warning: If IDA misidentifies the file format, the output will be garbage.

Mapping Reality: From Disk to Virtual Memory

IDA does not read the file linearly like a text editor. It performs Memory Mapping. It reads the executable and maps it into virtual memory exactly where it would reside during execution. The addresses you see in IDA are the specific addresses that would exist in live memory.



Pass 1: The Skeleton (Recursive Descent)



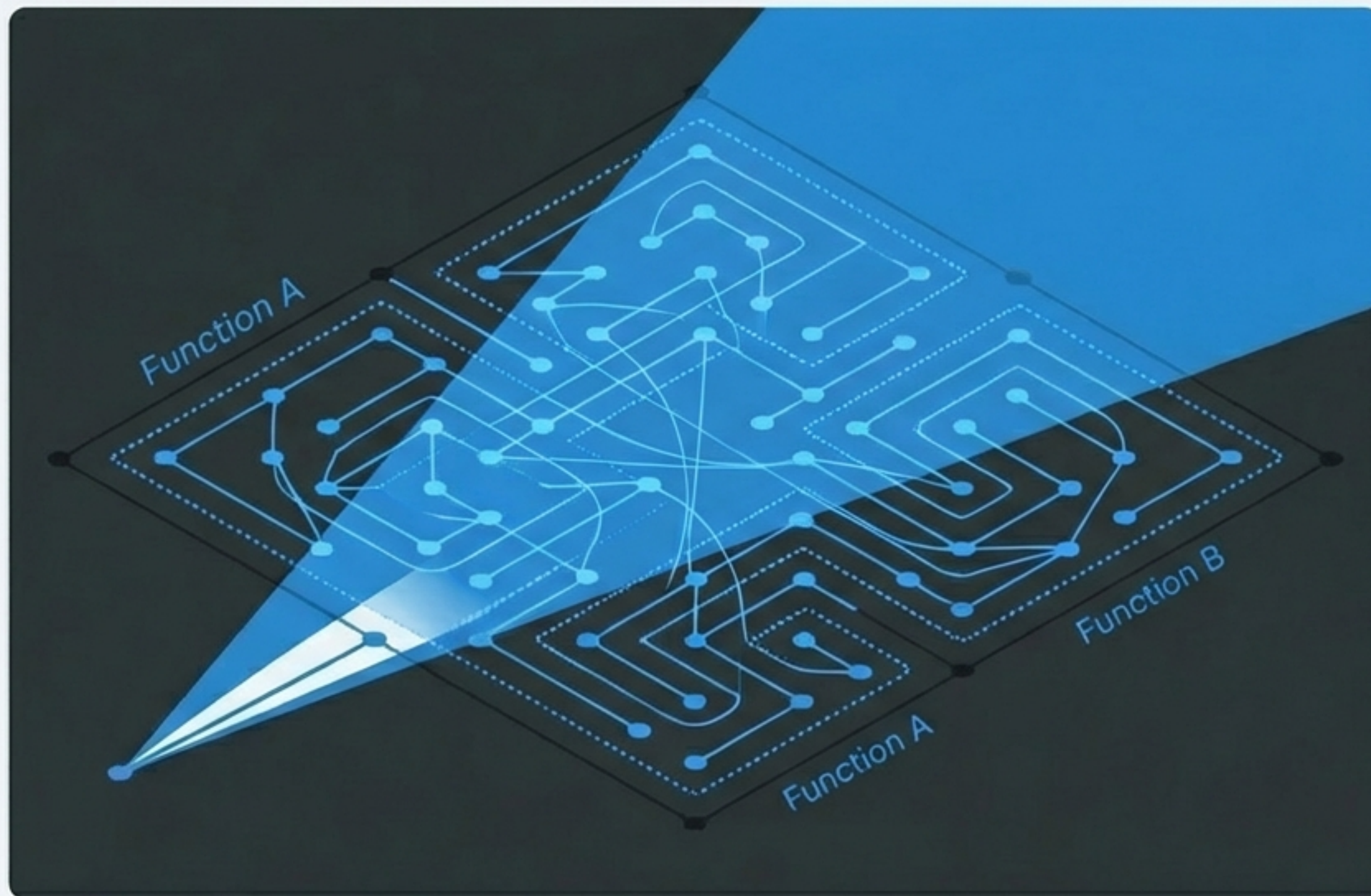
Mechanism: Recursive Descent Disassembly.

The Action:

- Starts at the Entry Point.
- Follows code flow linearly.
- Decodes raw bytes into mnemonics.

What is Revealed: Instruction types, operand lengths, and the distinction between code and data.

Pass 2: The Nervous System (Semantic Analysis)



Narrative: Once the skeleton is visible, IDA performs a second pass to add context and meaning.

Key Elements Added:

- **Cross-References (Xrefs):** The wires linking calls to destinations and jumps to labels.
- **Function Identification:** Grouping instructions into logical blocks.
- **Argument Detection:** Identifying parameters and local variables.

The Twin Worlds: Binary vs. Database

World A: Source

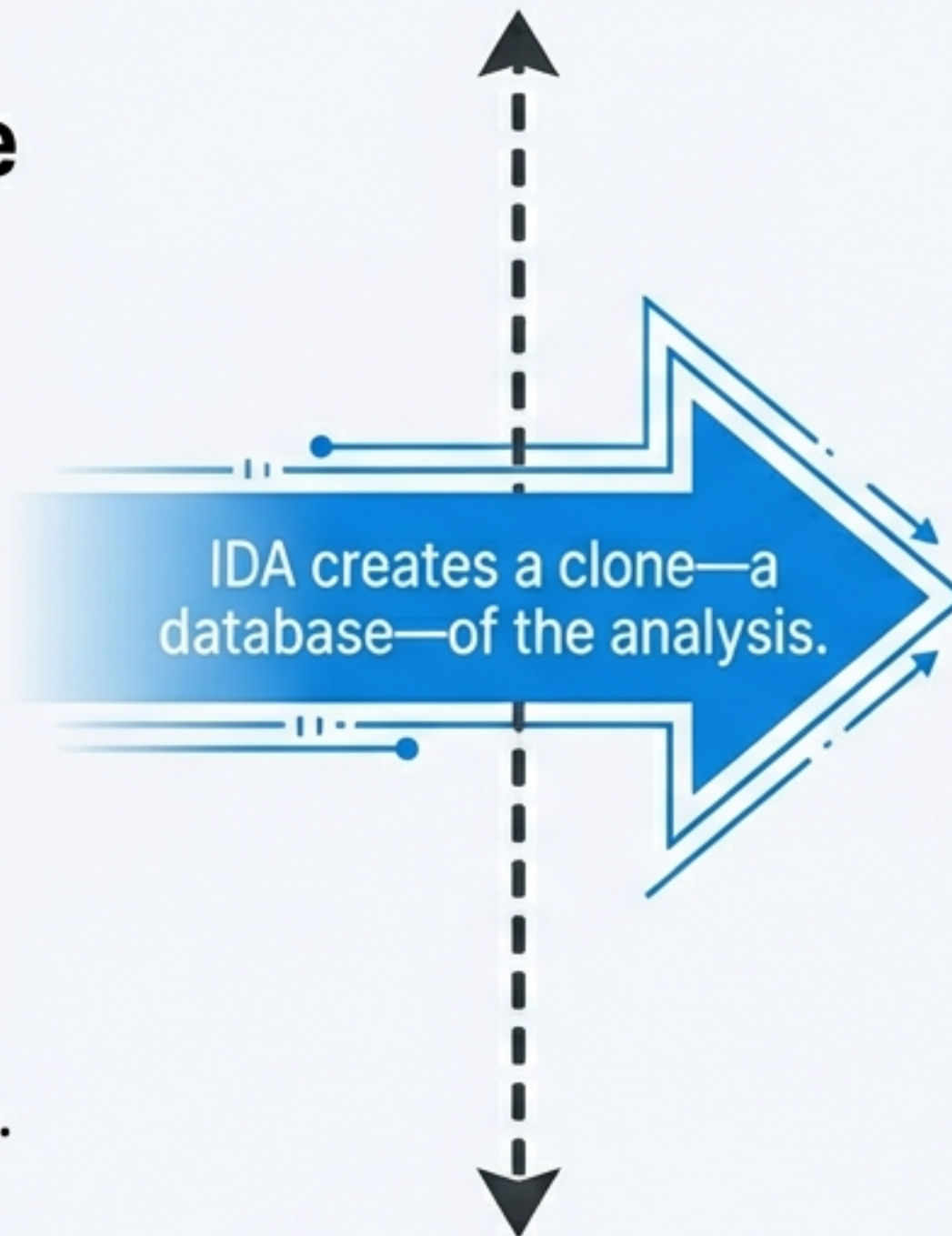


READ-ONLY.
The raw binary file.
IDA never touches this.

World B: Analysis

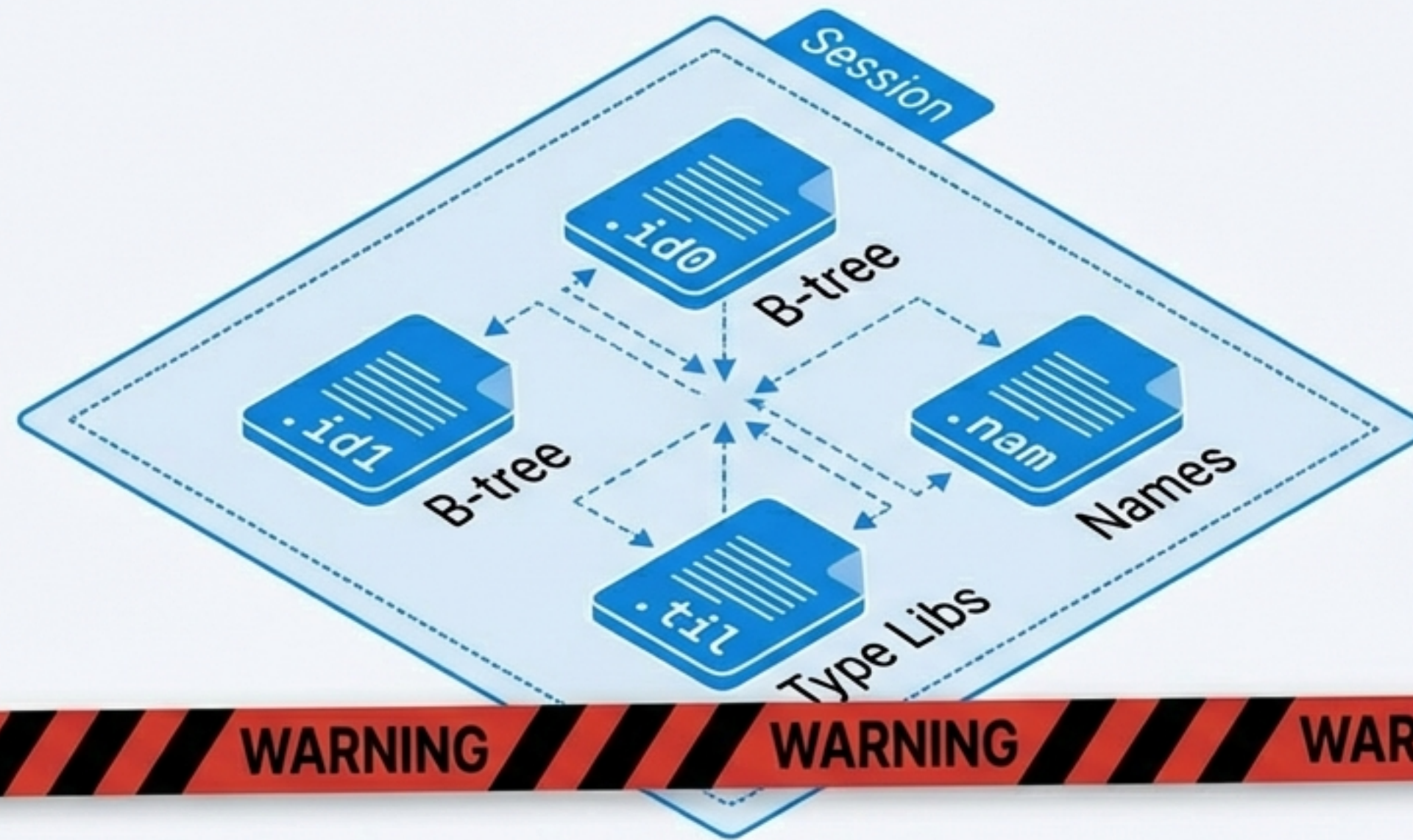


READ / WRITE.
The Analysis Database.
This is where your work lives.



All comments, colors, names, and labels are stored in the database, not the executable.

Inside the Vault: Temporary Files

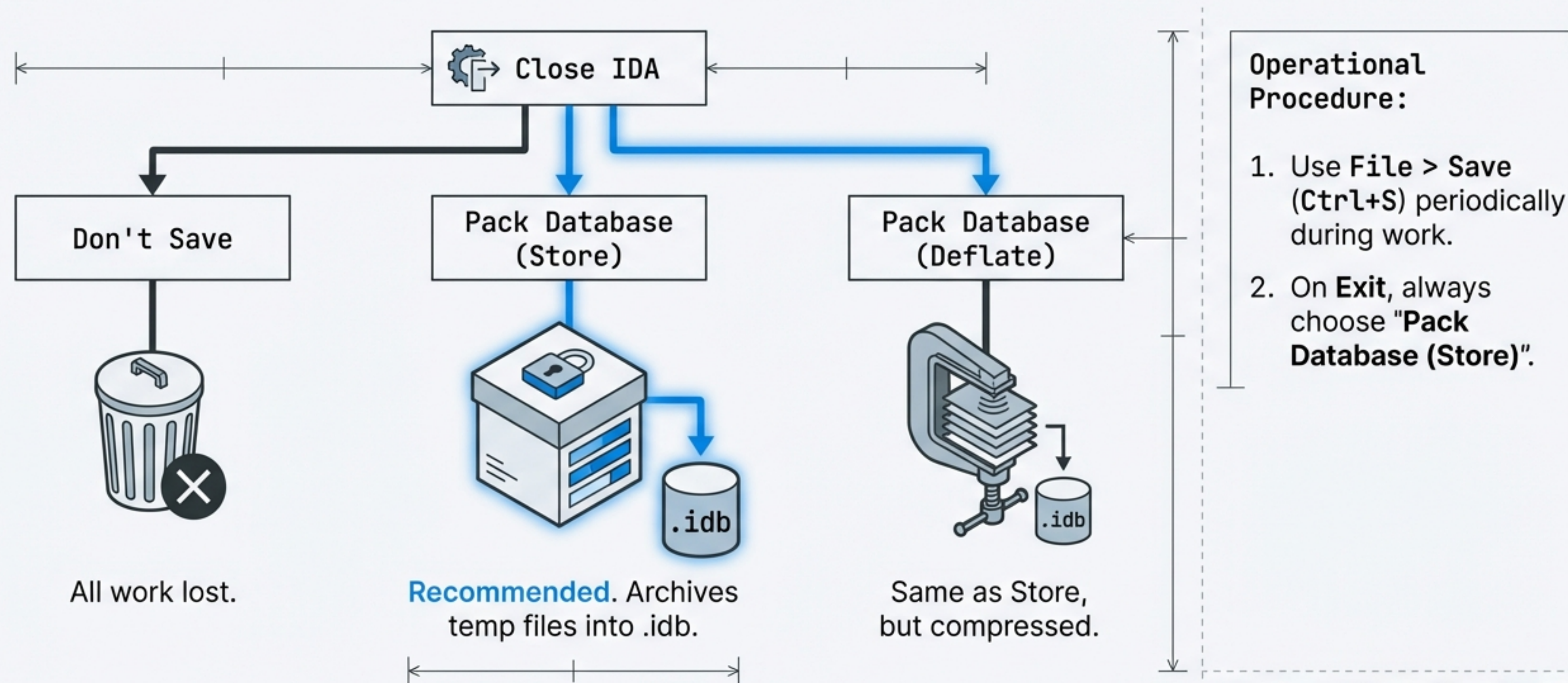


The **Hot Zone**: While IDA is running, the database is split into these active components in your directory.

WARNING: Never delete or rename these files while IDA is open. They are the active brain of your session. Doing so guarantees **corruption**.

Mission Critical: The Preservation Protocol

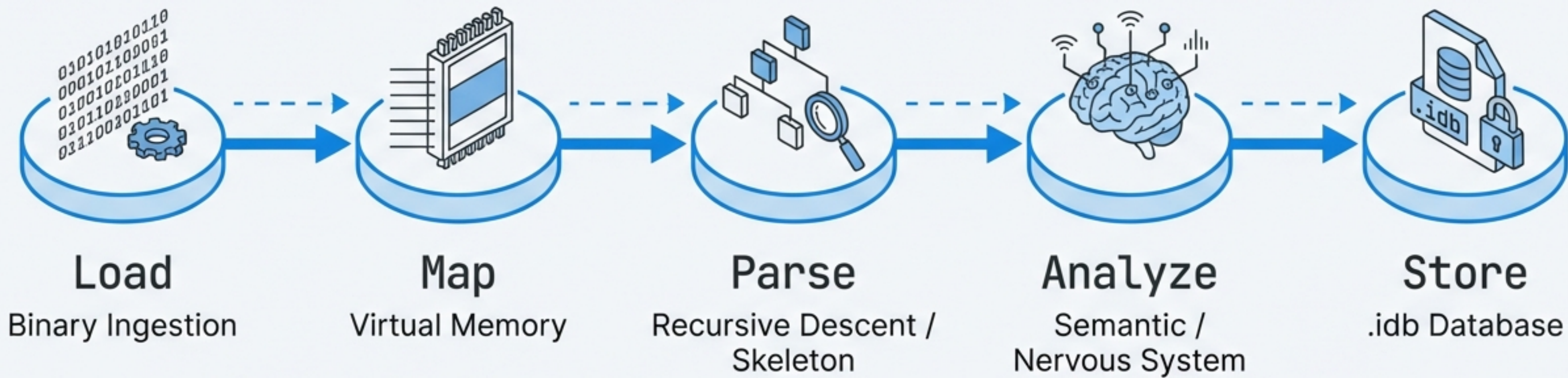
IDA Pro **does NOT autosave**. A crash means total data loss.



Operational Procedure:

1. Use **File > Save** (Ctrl+S) periodically during work.
2. On **Exit**, always choose "**Pack Database (Store)**".

The Architect's Blueprint



**"You are the architect. IDA is the blueprint.
Treat the database as your masterwork."**



Verify Your Mental Model

Why is the 'Demo' version unsuitable for this course?

It disables saving; you cannot submit work.

True or False:
Adding a comment in IDA modifies the original .exe file.

False. It modifies the database (.idb), not the binary.

What is the recommended action when closing IDA to ensure work is kept?

Pack database (Store) or Deflate.

